# A LOGARITHMIC SOBOLEV INEQUALITY FOR AN INTERACTING SPIN SYSTEM UNDER A GEOMETRIC REFERENCE MEASURE

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Logarithmic Sobolev inequalities are an essential tool in the study of interacting particle systems, cf. e.g. [4], [5]. In this note we show that the logarithmic Sobolev inequality proved on the configuration space $\mathbb{N}^{\mathbb{Z}^d}$ under Poisson reference measures in [1] can be extended to geometric reference measures using the results of [2]. As a corollary we obtain a deviation estimate for an interacting particle system.

## 1. Logarithmic Sobolev inequality for the geometric distribution

Consider the forward and backward gradient operators

$$\mathrm{d}^+ f(k) = f(k+1) - f(k), \qquad \mathrm{d}^- f(k) = 1_{\{k \geq 1\}}(f(k-1) - f(k)), \quad k \in \mathbb{N},$$

and the Laplacian

$$\mathscr{L} = -\mathrm{d}_\pi^{+*}\mathrm{d}^+ = \mathrm{d}^+ + \frac{1}{p}\mathrm{d}^-$$

which generates a Markov process on $\mathbb{N}$ whose invariant measure is the geometric distribution $\pi$ on $\mathbb{N}$ with parameter $p \in (0,1)$, i.e.

$$\pi(\{k\}) = (1-p)p^k, \qquad k \in \mathbb{N}.$$

Denote by $\mathbb{E}_\pi$ the expectation under $\pi$ and by $\mathrm{Ent}_\pi$ the entropy under $\pi$, defined as

$$\mathrm{Ent}_\pi[f] = \mathbb{E}_\pi[f \log f] - \mathbb{E}_\pi[f] \log \mathbb{E}_\pi[f].$$

We recall the modified logarithmic Sobolev inequality proved in [2] for the geometric distribution $\pi$.

**Theorem 1.1.** *Let $0 < c < -\log p$ and let $f : \mathbb{N} \to \mathbb{R}$ such that $|\mathrm{d}^+ f| \leq c$. We have*

$$\mathrm{Ent}_\pi \left[ e^f \right] \leq \frac{pe^c}{(1-p)(1-\sqrt{pe^c})} \mathbb{E}_\pi \left[ |\mathrm{d}^+ f|^2 e^f \right]. \tag{1.1}$$

In higher dimensions the multi-dimensional gradient is defined as

$$\mathrm{d}_i^+ f(k) = f(k + e_i) - f(k), \quad i = 1, \ldots, n,$$

where $f$ is a function on $\mathbb{N}^n$, $k = (k_1, \ldots, k_n) \in \mathbb{N}^n$, $(e_1, \ldots, e_n)$ is the canonical basis of $\mathbb{R}^n$, and the gradient norm is

$$\|\mathrm{d}^+ f(k)\|^2 = \sum_{i=1}^n |\mathrm{d}_i^+ f(k)|^2 = \sum_{i=1}^n |f(k + e_i) - f(k)|^2. \tag{1.2}$$

From the tensorization property of entropy, (1.1) still holds with respect to $\pi^{\otimes n}$ in any finite dimension $n$:

$$\mathrm{Ent}_{\pi^{\otimes n}} \left[ e^f \right] \leq \frac{pe^c}{(1-p)(1-\sqrt{pe^c})} \mathbb{E}_{\pi^{\otimes n}} \left[ \|\mathrm{d}^+ f\|^2 e^f \right], \tag{1.3}$$

provided $|d_i f| \leq c$, $i = 1, \ldots, n$. As a consequence the following deviation inequality for functions of several variables under $\pi^{\otimes n}$ has been proved in [2] using (1.1) and the Herbst method.

**Corollary 1.2.** *Let $0 < c < -\log p$ and let $f$ such that $|\mathrm{d}_i^+ f| \leq \beta$, $i = 1, \ldots, n$, and $\|\mathrm{d}^+ f\|^2 \leq \alpha^2$ for some $\alpha, \beta > 0$. Then for all $r > 0$,*

$$\pi^{\otimes n}(f - \mathbb{E}_{\pi^{\otimes n}}[f] \geq r) \leq \exp\left( - \min\left( \frac{c^2 r^2}{4 a_{p,c} \alpha^2 \beta^2}, \frac{rc}{\beta} - \alpha^2 a_{p,c} \right) \right), \tag{1.4}$$

*where*

$$a_{p,c} = \frac{pe^c}{(1-p)(1-\sqrt{pe^c})}$$

*denotes the logarithmic Sobolev constant in (1.1).*

Our goal in the next section will be to extend these results to interacting spin systems under a geometric reference measure.

## 2. Logarithmic Sobolev inequality for an interacting spin system

Given a bounded finite range interaction potential $\Phi = \{\Phi_R \ : \ R \subset \mathbb{Z}^d\}$, i.e.

$$\|\Phi\| = \sup_{k \in \mathbb{Z}^d} \sum_{R \ni k} \|\Phi_R\|_\infty < \infty,$$

let the Hamiltonian $H_\Lambda$ be defined as

$$H_\Lambda(\eta) = \sum_{R \bigcap \Lambda \neq \emptyset} \Phi_R(\eta_R),$$

where $\eta_R$ denotes the restriction of $\eta$ to $\mathbb{N}^R$, $R \subset \mathbb{Z}^d$. The Gibbs measure $\pi_\Lambda^\omega$ on $\mathbb{N}^\Lambda$ associated to a $\mathbb{N}$-valued spin system on a finite lattice $\Lambda \subset \mathbb{Z}^d$ with boundary condition $\omega \in \mathbb{N}^{\mathbb{Z}^d \setminus \Lambda}$ is defined by its density with respect to $\pi_\Lambda := \pi^{\otimes \Lambda}$ as:

$$\frac{d\pi_\Lambda^\omega}{d\pi_\Lambda}(\sigma) = \frac{1}{Z_\Lambda^\omega} e^{-H_\Lambda^\omega(\sigma)}, \quad \sigma \in \mathbb{N}^\Lambda,$$

where $\pi$ is the geometric reference distribution on $\mathbb{N}$, $Z_\Lambda^\omega$ is a normalization factor, and

$$H_\Lambda^\omega(\eta) = H_\Lambda(\eta_\Lambda \omega_{\Lambda^c}), \quad \eta \in \mathbb{N}^{\mathbb{Z}^d},$$

where $\eta_A \omega_B$ is defined as

$$(\eta_A \omega_B)_k = \eta_k 1_A(k) + \omega_k 1_B(k), \qquad k \in \mathbb{Z}^d,$$

whenever $\eta \in \mathbb{N}^A$, $\omega \in \mathbb{N}^B$, and $A, B \subset \mathbb{Z}^d$ are such that $A \cap B = \emptyset$. Let again

$$\mathrm{d}_k^+ f(\eta) = f(\eta + e_k) - f(\eta), \quad \text{and} \quad \mathrm{d}_k^- f(\eta) = 1_{\{\eta_k > 0\}} \left( f(\eta - e_k) - f(\eta) \right),$$

$\eta \in \mathbb{N}^{\mathbb{Z}^d}$, $k \in \mathbb{Z}^d$, for every function $f : \mathbb{N}^{\mathbb{Z}^d} \to \mathbb{R}$, where $(e_k)_{k \in \mathbb{Z}^d}$, denotes the canonical basis $\{e_k = 1_{\{k\}} \ : \ k \in \mathbb{Z}^d\}$. Consider the Markov generator

$$\mathscr{L}_\Lambda^\omega f(\eta) = \sum_{k \in \Lambda} \left( c_\Lambda^\omega(k, \eta, +) \mathrm{d}_k^+ f(\eta) + c_\Lambda^\omega(k, \eta, -) \mathrm{d}_k^- f(\eta) \right),$$

where $c_\Lambda^\omega(k, \eta, \pm)$ are rate functions such that $\mathscr{L}_\Lambda^\omega$ is self-adjoint in $L^2(\pi_\Lambda^\omega)$, i.e.

$$c_\Lambda^\omega(k, \eta, +) \pi_\Lambda^\omega(\{\eta\}) = c_\Lambda^\omega(k, \eta + e_k, -) \pi_\Lambda^\omega(\{\eta + e_k\}), \quad k \in \Lambda, \ \eta \in \mathbb{N}^\Lambda,$$

$$c_\Lambda^\omega(k, \eta, -) \pi_\Lambda^\omega(\{\eta\}) = c_\Lambda^\omega(k, \eta - e_k, +) \pi_\Lambda^\omega(\{\eta - e_k\}), \quad k \in \Lambda, \ \eta \in \mathbb{N}^\Lambda,$$

$\eta_k > 0$, cf. [1]. We assume that there exists a constant $C > 0$ depending on $\|\Phi\|$ only, with

$$\frac{1}{C} \leq c^\omega_\Lambda(k, \eta, +) \leq C, \qquad \eta \in \mathbb{N}^\Lambda, \ \Lambda \subset \mathbb{Z}^d, \ k \in \Lambda. \tag{2.1}$$

For $f : \mathbb{N}^{\mathbb{Z}^d} \to \mathbb{R}$ we let:

$$\mathscr{E}^\omega_\Lambda(e^f) = \sum_{k\in\Lambda} \int_{\mathbb{N}^{\mathbb{Z}^d}} c^\omega_\Lambda(k, \sigma, +) e^{f(\sigma)} |\mathrm{d}^+_k f(\sigma)|^2 d\pi^\omega_\Lambda(\sigma),$$

and

$$\mathscr{E}_\Lambda(e^f) = \sum_{k\in\Lambda} \int_{\mathbb{N}^{\mathbb{Z}^d}} e^{f(\sigma)} |\mathrm{d}^+_k f(\sigma)|^2 d\pi_\Lambda(\sigma).$$

Next we consider the family of rectangles of the form

$$R = R(k, l_1, ..., l_d) = k + ([1, l_1] \times \cdots \times [1, l_d]) \cap \mathbb{Z}^d,$$

where $k \in \mathbb{Z}^d$ and $l_1, \ldots, l_d \in \mathbb{N}$, with

$$\mathrm{size}(R) = \max_{k=1,\ldots,d} l_k.$$

Let $\mathscr{R}_L$ denote the set of rectangles such that

$$\mathrm{size}(R) \leq L \quad \text{and} \quad \mathrm{size}(R) \leq 10 \min_{k=1,\ldots,d} l_k.$$

**Definition 2.1.** We say that $\pi^\omega_\Lambda$ satisfies the mixing condition if there exists constants $C_1$ and $C_2$, depending on $d$ and $\|\Phi\|$ only, such that:

$$\sup_{\sigma,\omega} \left| \frac{\pi^\omega_\Lambda(\{\eta : \ \eta_A = \sigma_A\}) \pi^\omega_\Lambda(\{\eta : \ \eta_B = \sigma_B\})}{\pi^\omega_\Lambda(\{\eta \ : \ \eta_{A\cup B} = \sigma_{A\cup B}\})} - 1 \right| \leq C_1 e^{-C_2 d(A,B)}, \tag{2.2}$$

for all $L \geq 1$, $\Lambda \in \mathscr{R}_L$ and $A, B \subset \Lambda$ such that $A, B \in \mathscr{R}_L$ with $A \cap B = \emptyset$.

We refer to [1] and [4] for conditions on $\Phi$ under which (2.2) holds under a geometric reference measure.

Our goal is to prove the following logarithmic Sobolev inequality under the Gibbs measure $\pi^\omega_\Lambda$.

**Theorem 2.2.** *Assume that the mixing condition (2.2) holds, and let* $0 < c < -\log p$. *Then there exists a constant* $\gamma_c > 0$, *independent of* $\Lambda$ *and* $\omega$, *such that*

$$\mathrm{Ent}_{\pi^\omega_\Lambda}\left[e^f\right] \leq \gamma_c \mathscr{E}^\omega_\Lambda(e^f), \tag{2.3}$$

*for every* $f : \mathbb{N}^{\mathbb{Z}^d} \to \mathbb{R}$ *such that* $\|\mathrm{d}^+ f\|_{l^\infty(\Lambda)} \leq c$, $\pi_\Lambda$*-a.e.*

In particular we have

$$\mathrm{Ent}_{\pi_\Lambda^\omega}\left[e^f\right] \leq \gamma_c \left\| \sum_{k\in\Lambda} c_\Lambda^\omega(k,\cdot,+)|\mathrm{d}_k^+ f(\cdot)|^2 \right\|_{L^\infty(\pi_\Lambda)} \times \mathbb{E}_{\pi_\Lambda^\omega}\left[e^f\right],$$

which implies, as in Corollary 1.2, a deviation inequality under Gibbs measures.

**Corollary 2.3.** *Assume that the mixing condition (2.2) holds, and let* $0 < c < -\log p$. *Let* $f$ *be such that* $\|\mathrm{d}^+ f\|_{l^\infty(\Lambda)} \leq \beta$ *and*

$$\sum_{k\in\Lambda} c_\Lambda^\omega(k,\eta,+)|\mathrm{d}_k^+ f(\eta)|^2 \leq \alpha^2, \quad \pi_\Lambda(d\eta) - a.e., \tag{2.4}$$

*for some* $\alpha, \beta > 0$. *Then for all* $r > 0$,

$$\pi_\Lambda^\omega\left(f - \mathbb{E}_{\pi_\Lambda^\omega}[f] \geq r\right) \leq \exp\left(-\min\left(\frac{c^2 r^2}{4\gamma_c \alpha^2 \beta^2}, \frac{rc}{\beta} - \alpha^2\gamma_c\right)\right). \tag{2.5}$$

Due to Hypothesis (2.1), condition (2.4) can be replaced by

$$\|\mathrm{d}^+ f(\eta)\|^2_{l^2(\Lambda)} \leq C^{-1}\alpha^2, \quad \pi_\Lambda(d\eta) - a.e.$$

Denoting by $\Pi$ denote the infinite volume Gibbs measure associated to $\pi_\Lambda^\omega$, for some $r_0 > 0$ we get the Ruelle type bound:

$$\Pi(\{\eta \in \mathbb{N}^{\mathbb{Z}^d} : |\eta_\Lambda| \geq r|\Lambda|\}) \leq \exp\left(-(cr - C\gamma_c)|\Lambda| + c\mathbb{E}_\Pi[|\eta_\Lambda|]\right), \quad r > r_0,$$

for all finite subset $\Lambda$ of $\mathbb{Z}^d$, under the mixing condition (2.2). Indeed, it suffices to apply the uniform bound (2.5) with $f(\eta) = |\eta_\Lambda|$, $\alpha^2 = C|\Lambda|$, $\beta = 1$, and the compatibility condition

$$\Pi(E) = \int_{\mathbb{N}^{\mathbb{Z}^d}} \pi_\Lambda^\omega(E_\Lambda)\Pi(d\omega),$$

to $E = \{\eta \in \mathbb{N}^{\mathbb{Z}^d} : |\eta_\Lambda| \geq r|\Lambda|\}$, with

$$E_\Lambda := \{\eta \in \mathbb{N}^\Lambda : \eta_\Lambda \omega_{\Lambda^c} \in E\} = \{\eta \in \mathbb{N}^\Lambda : |\eta_\Lambda| \geq r|\Lambda|\}.$$

This shows in particular that $\Pi$ satisfies the $(RPB)^1$ condition in [3].

## 3. Proof of Theorem 2.2

Recall that for $0 < c < -\log p$, by tensorization, Theorem 1.1 yields as in (1.3) the logarithmic Sobolev inequality

$$\mathrm{Ent}_{\pi_\Lambda}[e^f] \leq s_c \mathscr{E}_\Lambda(e^f), \tag{3.1}$$

for all $f : \mathbb{N}^{\mathbb{Z}^d} \to \mathbb{R}$ such that $\|\mathrm{d}^+ f\|_{l^\infty(\Lambda)} \leq c$, $\pi_\Lambda$-a.e., with an optimal constant $s_c \leq a_{p,c}$ which is independent of $\Lambda \subset \mathbb{Z}^d$. Let now $s_{\Lambda,\omega,c}$ denote the optimal constant in the inequality

$$\mathrm{Ent}_{\pi^\omega_\Lambda}[e^f] \leq s_{\Lambda,\omega,c} \mathscr{E}^\omega_\Lambda(e^f), \qquad \|\mathrm{d}^+ f\|_{l^\infty(\Lambda)} \leq c.$$

**Lemma 3.1.** *For every $\Lambda \subset \mathbb{Z}^d$, there exists a constant $A := Ce^{4|\Lambda|\|\Phi\|} > 0$ depending only on $|\Lambda|$, $c$ and independent of $\omega \in \mathbb{N}^{\mathbb{Z}^d \setminus \Lambda}$, such that*

$$\frac{s_c}{A} \leq s_{\Lambda,\omega,c} \leq A\, s_c.$$

*Proof.* We follow the proof of Proposition 3.1 in [1]. From (2.1) we obtain:

$$C^{-1} e^{-2|\Lambda|\|\Phi\|} \mathscr{E}_\Lambda(e^f) \leq \mathscr{E}^\omega_\Lambda(e^f) \leq Ce^{2|\Lambda|\|\Phi\|} \mathscr{E}_\Lambda(e^f). \tag{3.2}$$

From the relation

$$\mathrm{Ent}_\mu[f] = \min_{t>0} \mathbb{E}_\mu[f \log f - f \log t - f + t]$$

and the bound

$$e^{-2|\Lambda|\|\Phi\|} \leq \frac{d\pi^\omega_\Lambda}{d\pi_\Lambda} \leq e^{2|\Lambda|\|\Phi\|},$$

we have

$$e^{-2|\Lambda|\|\Phi\|} \mathrm{Ent}_{\pi_\Lambda}\left[e^f\right] \leq \mathrm{Ent}_{\pi^\omega_\Lambda}\left[e^f\right] \leq e^{2|\Lambda|\|\Phi\|} \mathrm{Ent}_{\pi_\Lambda}\left[e^f\right],$$

from which the conclusion follows using (3.1) and (3.2). □

Let for $L \geq 1$:

$$S_{L,c} := \sup_{R \in \mathscr{R}_L} \sup_{\omega \in \mathbb{N}^{\mathbb{Z}^d \setminus \Lambda}} s_{R,\omega,c} \leq Cs_c e^{4|\Lambda|\|\Phi\|} < \infty,$$

which is finite by Lemma 3.1.

**Proposition 3.2.** *Assume the mixing condition (2.2) is satisfied. Then there exists a constant $\kappa$ depending on $\|\Phi\|$, such that*

$$S_{2L,c} \leq \left(1 - \frac{\kappa}{\sqrt{L}}\right)^{-1} S_{L,c} \tag{3.3}$$

*for $L$ large enough.*

*Proof.* The proof of this proposition is identical to that of Proposition 4.1, pp. 1970-1972 and Proposition 5.1, p. 1975 in [1], replacing the Dirichlet form used in [1] with $\mathscr{E}^\omega_\Lambda$. □

Finally, Theorem 2.2 is proved by taking $\gamma_c = \sup_L S_{L,c}$, which is finite from Proposition 3.2.

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